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Tractor with exhaust-gas cleaning devices

Abstract

A tractor includes: a vehicle body; an engine bonnet forming an engine compartment; an engine; a first exhaust-gas cleaning device that purifies an exhaust gas and is above the engine such that its long side is substantially parallel to a vehicle-body lateral width direction; a second exhaust-gas cleaning device that purifies the exhaust gas and is behind the engine such that its long side is substantially parallel to the lateral width direction; a vehicle body frame; and a supporting tool that is supported by the vehicle body frame, supports the first and second exhaust-gas cleaning devices, and includes a support column that extends upward from the vehicle body frame along a front portion of the second exhaust-gas cleaning device, and a support arm that extends forward from an upper portion of the support column along a lower portion of the first exhaust-gas cleaning device.

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Background/Summary

BACKGROUND

Technical Field

(1) The present invention relates to a tractor.

Description of Related Art

(2) As shown in patent literature 1 and 2, for example, the tractor is equipped with an engine bonnet (bonnet) that forms an engine compartment, and in the engine compartment, an exhaust-gas cleaning device (exhaust-gas cleaning process device) is installed above the engine.

PRIOR ART LITERATURE

(3) [Patent Literature 1] JP 2013-112284 A

(4) [Patent Literature 2] JP 2015-143508 A

(5) There is a demand for a tractor to be provided with a plurality of exhaust-gas cleaning devices for cleaning exhaust gas discharged from an engine. Examples of the plurality of exhaust-gas cleaning devices include an exhaust-gas cleaning device that cleans exhaust gas using a cleaning filter and an exhaust-gas cleaning device that cleans exhaust gas using a reducing agent.

(6) When the conventional art is adopted and the first exhaust-gas cleaning device and the second exhaust-gas cleaning device are installed in the engine compartment, the first exhaust-gas cleaning device and the second exhaust-gas cleaning device are located above the engine, and thus, even when the size of one of the first exhaust-gas cleaning device and the second exhaust-gas cleaning device is smaller than the size of the other exhaust-gas cleaning device, the mounting height of the engine bonnet must be set to correspond to the larger exhaust-gas cleaning device. Therefore, the mounting height of the engine bonnet is increased.

SUMMARY

(7) One or more embodiments of the present invention provide a tractor for which it is possible to provide a first exhaust-gas cleaning device and a second exhaust-gas cleaning device in an engine compartment while keeping the mounting height of the engine bonnet low, keeping the front-back length of the engine compartment short, and supporting the first exhaust-gas cleaning device and the second exhaust-gas cleaning device onto the vehicle body frame while in an appropriate connection state.

(8) The tractor of one or more embodiments of the present invention is provided with: a vehicle body; an engine bonnet forming an engine compartment; an engine that is disposed in the engine compartment; a first exhaust-gas cleaning device and a second exhaust-gas cleaning device provided in the engine compartment for purifying exhaust gas discharged from the engine; and a support member (i.e., a supporting tool) supported by the vehicle body frame and supporting the first exhaust-gas cleaning device and the second exhaust-gas cleaning device is provided, the first exhaust-gas cleaning device being provided above the engine and having a long side substantially parallel to a vehicle-body lateral width direction (or a vehicle-body horizontal direction), and the second exhaust-gas cleaning being provided behind the engine and having a long side substantially parallel to the vehicle-body lateral width direction; wherein, the supporting member is provided with a support column extending upward from the vehicle body frame along the front portion of the second exhaust-gas cleaning device while supporting the front portion and a support arm extending forward from the upper portion of the support column along a lower portion of the first exhaust-gas cleaning device while supporting the lower portion.

(9) According to this configuration, the first exhaust-gas cleaning device, which is the smaller of the first exhaust-gas cleaning device and the second exhaust-gas cleaning device, is installed above the engine, and the larger second exhaust-gas cleaning device is installed behind the engine, and the mounting height of the engine bonnet can be set based on the size of the small first exhaust-gas cleaning device. Furthermore, since the long sides of both the first and second exhaust-gas cleaning devices are substantially parallel to the vehicle-body lateral width direction, the first and second exhaust-gas cleaning devices can be accommodated compared to the case where the long sides are substantially parallel to the vehicle-body longitudinal direction, and it is possible to shorten the length of the space in a vehicle-body longitudinal direction (or a vehicle-body front-back direction) necessary for the installation of the first exhaust-gas cleaning device, while keeping the mounting height of the engine bonnet low and the longitudinal length of the engine compartment shorter, and a second exhaust-gas cleaning device can be provided in the engine compartment.

(10) Because the positional relationship between the first exhaust-gas cleaning device and the second exhaust-gas cleaning device is supported by the vehicle body frame in a state where the positional relationship is set by the support member, the first exhaust-gas cleaning device and the

second exhaust-gas cleaning device can be easily connected; for example, even in a connection structure that does not have a function enabling adjustment of the positional relationship between the first exhaust-gas cleaning device and the second exhaust-gas cleaning device, the first exhaust-gas cleaning device and the second exhaust-gas cleaning device can be connected in a positional relationship advantageous for connection.

(11) In one or more embodiments of the present invention, the vehicle body frame is provided with a flywheel housing extending backward from the back portion of the engine and positioned below the second exhaust-gas cleaning device, and the support column extends upward from the flywheel housing toward the vehicle body.

(12) According to this configuration, since the support column extends upward from the flywheel housing located below the second exhaust-gas cleaning device, the length of the support column can be shortened, and it is possible to firmly support the first exhaust-gas cleaning device and the second exhaust-gas cleaning device onto the vehicle body frame such that the first exhaust-gas cleaning device and the second exhaust-gas cleaning device do not readily swing.

(13) The tractor of one or more embodiments of the present invention is provided with a position adjustment tool that can adjust a support position of the second exhaust-gas cleaning device on the support column in a vehicle-body vertical direction (or a vehicle-body up-down direction), the vehicle-body longitudinal direction, and the vehicle-body lateral width direction.

(14) According to the present configuration, even when there is a positional shift between the first exhaust-gas cleaning device and the second exhaust-gas cleaning device in the vehicle-body vertical direction, the vehicle-body longitudinal direction, and the vehicle-body lateral width direction due to manufacturing errors in the support member, by for example, adjusting the positional shift via the position adjustment tool, for example, the first exhaust-gas cleaning device and the second exhaust-gas cleaning device can be connected even in a connection structure not provided with a function for correcting the positional shift, and thus the connection structure can be simplified in that a special positional shift adjustment portion is not required in a connection structure that connects the first exhaust-gas cleaning device and the second exhaust-gas cleaning device.

(15) The tractor of one or more embodiments of the present invention is provided with a lower support portion that is supported below the support column and supports a lower portion of the second exhaust-gas cleaning device.

(16) According to this configuration, the front portion of the second exhaust-gas cleaning device is supported by the support column, and the lower portion of the second exhaust-gas cleaning device is supported by the support column, and thus the second exhaust-gas cleaning device can be firmly supported by the support member.

(17) The tractor of one or more embodiments of the present invention is provided with a second position adjustment tool that can adjust a support position of the second exhaust-gas cleaning device on the lower support portion in the vehicle-body vertical direction, the vehicle-body longitudinal direction, and the vehicle-body lateral width direction.

(18) According to the present configuration, even when there is a positional shift between the first exhaust-gas cleaning device and the second exhaust-gas cleaning device in the vehicle-body vertical direction, the vehicle-body longitudinal direction, and the vehicle-body lateral width direction due to manufacturing errors in the support member, for example, by adjusting the positional shift via a second position adjustment tool, for example, the first exhaust-gas cleaning device and the second exhaust-gas cleaning device can be connected even in a connection structure not provided with a function for correcting the positional shift, and thus the connection structure can be simplified in that a special positional shift adjustment portion is not required in a connection structure that connects the first exhaust-gas cleaning device and the second exhaust-gas cleaning device.

(19) In one or more embodiments of the present invention, the lower portion of the support column

comprises a washer portion that is connected to the vehicle body frame to make the support column supported by the vehicle body frame, and the lower support portion constitutes a part of the washer portion.

(20) According to this configuration, the washer portion serves as a connection member for supporting the lower support portion on the support column, and thus the structure of providing the lower support portion on the support post portion can be simplified without requiring a special connection member.

Description

BRIEF DESCRIPTION OF DRAWINGS

- (1) FIG. 1 is a left side view illustrating an entirety of a tractor.
- (2) FIG. 2 is a left side view illustrating a first exhaust-gas cleaning device and a second exhaust-gas cleaning device.
- (3) FIG. 3 is a plan view illustrating the first exhaust-gas cleaning device and the second exhaust-gas cleaning device.
- (4) FIG. 4 is a back view illustrating a support member.
- (5) FIG. 5 is a perspective view of the support member in an exploded state.
- (6) FIG. 6 is a right side view illustrating left and right brake operation devices.
- (7) FIG. 7 is an explanatory diagram illustrating a summary of assembly of the brake pedal.
- (8) FIG. 8 is a back view illustrating a steering-post cover and a panel cover.
- (9) FIG. 9 is a back view of a panel cover provided with another example of one or more embodiments.
- (10) FIG. 10 is a perspective view of a steering-post cover provided with another example of one or more embodiments.

DETAILED DESCRIPTION OF EMBODIMENTS

- (11) Embodiments of the present invention are described below based on the drawings.
- (12) Note that in the following description, in regards to a traveling vehicle body of a tractor, the direction of arrow F illustrated in FIG. 1 is defined as “vehicle-body front”, the direction of arrow B is defined as “vehicle-body back”, the direction of arrow U is defined as “vehicle-body up”, the direction of arrow D is defined as “vehicle-body down”, the direction heading toward the surface of the page is defined as “vehicle-body left”, and the direction heading toward the reverse face of the page is defined as “vehicle-body right”.
- (13) [Overall Configuration of Tractor]
- (14) As illustrated in FIG. 1, the tractor is provided with a traveling vehicle body 3 supported by a pair of left and right front wheels 1, which can be steered and driven, and a pair of left and right back wheels 2, which can be driven. A vehicle body frame 4 of the traveling vehicle body 3 is constituted by an engine 5, a flywheel housing 6 connected to a back portion of the engine 5, a clutch housing 7 connected to a back portion of the flywheel housing 6, a transmission case 7a connected to a back portion of the clutch housing 7, and a front frame 8 connected to a lower portion of the engine 5. A motor unit 9 provided with the engine 5 is formed in a front portion of the traveling vehicle body 3. A driver's seat 10 and a driving unit 12, which is provided with a steering wheel 11 whereby a steering operation of the front wheels 1 is performed, are formed in a back portion of the traveling vehicle body 3. A linking mechanism 13, which connects a work apparatus such as a rotary tilling apparatus (not illustrated) in a manner enabling a raising and lowering operation of the work apparatus, and a power takeoff shaft 14, which takes power from the engine 5 and outputs this to the connected work apparatus, are provided in a back portion of the transmission case 7a. Reference numeral 20 illustrated in FIG. 1 is a ROPS frame.
- (15) [Configuration of Motor Unit]

(16) As illustrated in FIGS. 1 and 2, the motor unit 9 is provided with an engine compartment 15. The engine compartment 15 is formed by an engine bonnet 16, which covers the engine compartment 15 from above and the front; a partition member 17 that forms a partition between the engine compartment 15 and the driving unit 12; and the like.

(17) As illustrated in FIGS. 1 and 2, the engine 5, a radiator 18 that cools the engine 5, and a first exhaust-gas cleaning device (DPF) 21 and second exhaust-gas cleaning device (SCR) 22 that perform cleaning processes of exhaust gas exhausted by the engine 5 are provided in the engine compartment 15.

(18) [Configuration of Radiator]

(19) As illustrated in FIG. 1, the radiator 18 is provided in front of the engine 5. A blowing action of a rotary fan 19 positioned between the radiator 18 and the engine 5 introduces cooling air from outside the engine compartment 15 to inside the engine compartment and supplies the cooling air to the radiator 18. The cooling air is supplied to the radiator 18 in a state of the cooling air passing through the radiator 18 from front to back. In the radiator 18, engine cooling water is cooled by heat exchange between the supplied cooling air and the engine cooling water. The engine 5 is cooled by the cooled engine cooling water being supplied to the engine 5.

(20) [Configurations of First Exhaust-Gas Cleaning Device and Second Exhaust-Gas Cleaning Device]

(21) The engine 5 is a diesel engine. As illustrated in FIGS. 2 and 3, the first exhaust-gas cleaning device 21 is provided above the engine 5 in a state wherein a long side of the first exhaust-gas cleaning device is substantially parallel to a vehicle-body lateral width direction. A maximum length H in a vehicle-body vertical direction of the first exhaust-gas cleaning device 21 is set to be shorter than a maximum length W in a vehicle-body longitudinal direction of the first exhaust-gas cleaning device 21. As illustrated in FIGS. 2 and 3, an exhaust-gas suctioning portion 21a, which is provided to a part on one end side, in a vehicle-body lateral width direction, of the first exhaust-gas cleaning device 21, and an exhaust-gas exhausting portion 5a, which is provided to the engine 5, are connected. An exhaust-gas discharging portion 21b is provided to a part on another end side, in the vehicle-body lateral width direction, of the first exhaust-gas cleaning device 21. In one or more embodiments, the exhaust-gas suctioning portion 21a protrudes downward from a lower portion of the first exhaust-gas cleaning device 21 at an end portion on a vehicle-body left horizontal side of the first exhaust-gas cleaning device 21, and the exhaust-gas discharging portion 21b protrudes backward from an end portion on a vehicle-body right horizontal side of the first exhaust-gas cleaning device 21.

(22) In the first exhaust-gas cleaning device 21, the exhaust gas exhausted by the engine 5 from the exhaust-gas exhausting portion 5a is suctioned by the exhaust-gas suctioning portion 21a into the device, and diesel microparticles included in the suctioned exhaust gas are collected by a collection filter (not illustrated). This performs an exhaust-gas cleaning process of decreasing the diesel microparticles. The exhaust gas subjected to the cleaning process is discharged from the exhaust-gas discharging portion 21b.

(23) As illustrated in FIGS. 2 and 3, the second exhaust-gas cleaning device 22 is provided behind the engine 5 in a state wherein a long side of the second exhaust-gas cleaning device is substantially parallel to the vehicle-body lateral width direction. An exhaust-gas introduction portion 22a, which is provided to a part on one end side, in the vehicle-body lateral width direction, of the second exhaust-gas cleaning device 22, and the exhaust-gas discharging portion 21b of the first exhaust-gas cleaning device 21 are connected by a connecting pipe 23. An exhaust-gas discharging portion 22b is provided to a part on another end side, in the vehicle-body lateral width direction, of the second exhaust-gas cleaning device 22. In one or more embodiments, the exhaust-gas introduction portion 22a is provided to an end portion on a vehicle-body left horizontal side of the second exhaust-gas cleaning device 22, and the exhaust-gas discharging portion 22b is provided to an end portion on a vehicle-body right horizontal side of the second exhaust-gas cleaning device

22.

(24) In the second exhaust-gas cleaning device **22**, the exhaust gas discharged by the first exhaust-gas cleaning device **21** from the exhaust-gas discharging portion **21b** is supplied by the connecting pipe **23** to the exhaust-gas introduction portion **22a** and introduced by the exhaust-gas introduction portion **22a** into the device, and the introduced exhaust gas is subjected to the cleaning process by a reducing agent. Specifically, aqueous urea as the reducing agent is injected into the introduced exhaust gas, hydrolyzing the exhaust gas. This performs an exhaust-gas cleaning process of decreasing nitrogen oxides included in the exhaust gas. The exhaust gas subjected to the cleaning process is exhausted from the exhaust-gas discharging portion **22b** to a vehicle-body horizontal outer side.

(25) As illustrated in FIG. **3**, the first exhaust-gas cleaning device **21** is supported in an attachment disposition wherein the part on the other end side, in the vehicle-body lateral width direction, is swung and displaced toward a vehicle-body front side with respect to the part on the one end side, in the vehicle-body lateral width direction, in a state wherein the exhaust-gas suctioning portion **21a** is a swinging shaft. That is, the first exhaust-gas cleaning device **21** is supported in a disposition wherein in a plan view, the longitudinal direction thereof is inclined relative to the vehicle-body lateral width direction. As illustrated in FIG. **2**, the exhaust-gas introduction portion **22a** of the second exhaust-gas cleaning device **22** extends diagonally backward and upward from the second exhaust-gas cleaning device **22**. A positional relationship between the exhaust-gas discharging portion **21b** of the first exhaust-gas cleaning device **21** and the exhaust-gas introduction portion **22a** of the second exhaust-gas cleaning device **22** can be made to be a positional relationship appropriate for adopting a connecting pipe **23** having no bellows or other adjustment means of adjusting the positional relationship between the exhaust-gas discharging portion **21b** and the exhaust-gas introduction portion **22a**. This can be done by bringing the first exhaust-gas cleaning device **21** and the second exhaust-gas cleaning device **22** in proximity to each other in the vehicle-body longitudinal direction and by directly connecting the exhaust-gas suctioning portion **21a** of the first exhaust-gas cleaning device **21** to the exhaust-gas exhausting portion **5a** of the engine **5**.

(26) As illustrated in FIGS. **2** and **3**, a back portion **21c** of the first exhaust-gas cleaning device **21** is positioned on a vehicle-body back side of a back end portion of the engine **5**. In a plan view, the back portion **21c** of the first exhaust-gas cleaning device **21** and the second exhaust-gas cleaning device **22** overlap.

(27) As illustrated in FIGS. **2** and **3**, the partition member **17** that forms the partition between the engine compartment **15** and the driving unit **12** is provided across a back location of the second exhaust-gas cleaning device **22** and a lower location of the second exhaust-gas cleaning device **22**. The partition member **17** is configured so a part **17a**—positioned in the lower location of the second exhaust-gas cleaning device **22**—of the partition member **17** is positioned further toward a vehicle-body front side than a part **17b**—positioned in the back location of the second exhaust-gas cleaning device **22**—of the partition member **17**. The part **17a**—positioned in the lower location of the second exhaust-gas cleaning device **22**—of the partition member **17** is provided with a portion in an inclined state that, in moving toward its lower end side, is positioned more to a vehicle-body front side. A footwell of the driving unit **12** can be extended below the second exhaust-gas cleaning device **22**.

(28) As illustrated in FIG. **6**, electrical wires **24** are provided across the engine compartment **15** and the driving unit **12**. In one or more embodiments, a plurality of electrical wires **24** is provided. However, it is also possible for only one to be provided. The electrical wires **24** enter from the engine compartment **15** into the footwell of the driving unit **12** by passing below a part positioned in a central portion in the vehicle-body lateral width direction of the partition member **17**. The electrical wires **24** transmit information relating to actuation of the engine **5** to gauges of the driving unit **12**. A portion of the electrical wires **24** that is positioned on a driving-unit side of the

partition member **17** is wired in a state of conforming to the part **17a**—positioned in the lower location of the second exhaust-gas cleaning device **22**—of the partition member **17** in the footwell of the driving unit **12**. The electrical wires **24** pass through a conduit **25**. The conduit **25** is supported on the partition member **17** by a clamp **26**.

(29) [Support Member of Exhaust-Gas Cleaning Devices]

(30) As illustrated in FIGS. **2**, **3**, and **4**, the first exhaust-gas cleaning device **21** and the second exhaust-gas cleaning device **22** are supported on the vehicle body frame **4** via a support member (i.e., the supporting tool) **30**. The first exhaust-gas cleaning device **21** and the second exhaust-gas cleaning device **22** are supported by the vehicle body frame **4** in a state wherein the positional relationship between the exhaust-gas discharging portion **21b** of the first exhaust-gas cleaning device **21** and the exhaust-gas introduction portion **22a** of the second exhaust-gas cleaning device **22** is set by the support member **30**.

(31) As illustrated in FIGS. **2**, **3**, **4**, and **5**, the support member **30** is provided with a support column **31** that extends upward from the vehicle body frame **4** in a state of conforming to a front portion **22c** of the second exhaust-gas cleaning device **22** and supports the front portion **22c** of the second exhaust-gas cleaning device **22**, a support arm **32** that extends in a vehicle-body forward direction from an upper portion of the support column **31** in a state of conforming to a lower portion **21d** of the first exhaust-gas cleaning device **21** and supports the lower portion **21d** of the first exhaust-gas cleaning device **21**, and a lower support portion **33** that is provided to a lower portion of the support column **31** and supports a lower portion **22d** of the second exhaust-gas cleaning device **22**.

(32) As illustrated in FIGS. **2**, **4**, and **5**, the support column **31** has a washer portion **34** provided to the lower portion of the support column **31** and is supported on the vehicle body frame **4** by the washer portion **34** being connected to the vehicle body frame **4**. The connection of the washer portion **34** to the vehicle body frame **4** is performed by the flywheel housing **6** constituting the vehicle body frame **4**. The flywheel housing **6** is positioned below the second exhaust-gas cleaning device **22** and can keep a length of the support column **31** short.

(33) As illustrated in FIGS. **2**, **4**, and **5**, the support column **31** is provided with left and right strut rods **31a** lined up at an interval in the vehicle-body lateral width direction. A lower strut-portion reinforcing rod **31b**, which connects lower portions of the strut rods **31a**, and an upper strut-portion reinforcing rod **31c**, which connects upper portions of the strut rods **31a**, are provided across the left and right strut rods **31a**. The left and right strut rods **31a** are constituted by steel pipes.

(34) As illustrated in FIGS. **2**, **4**, and **5**, a front support portion **31d** supporting the second exhaust-gas cleaning device **22** is provided to each of the left and right strut rods **31a**. The front portion **22c** of the second exhaust-gas cleaning device **22** is supported by the support column **31** by front connecting portions **22e**—provided in two locations, left and right, in the front portion **22c** of the second exhaust-gas cleaning device **22**—being connected to the front support portion **31d**.

(35) As illustrated in FIGS. **2**, **4**, and **5**, washer plates **34a** provided to respective lower portions of the left and right strut rods **31a** are provided to the washer portion **34**. In the washer portion **34**, the left and right washer plates **34a** being connected to the flywheel housing **6** by a plurality of connecting bolts provides a detachable connection to the flywheel housing **6**.

(36) As illustrated in FIGS. **2**, **4**, and **5**, the lower support portion **33** is provided with lower support pieces **33a** provided to the lower portions of the left and right strut rods **31a**. The left and right lower support pieces **33a** are provided to the strut rods **31a** by being formed on the washer plates **34a**. The lower support portion **33** is provided to the washer portion **34**. The lower portion **22d** of the second exhaust-gas cleaning device **22** is supported by the lower support portion **33** by lower connecting portions **22f**—provided in two locations, left and right, in the lower portion **22d** of the second exhaust-gas cleaning device **22**—being connected to the lower support pieces **33a**.

(37) As illustrated in FIGS. **2**, **4**, and **5**, the support arm **32** is provided with arm bodies **32a** that extend in the vehicle-body forward direction from respective upper portions of the left and right

strut rods **31a**. An arm reinforcing rod **32b** connecting the left and right arm bodies **32a** is provided across distal end portions of the left and right arm bodies **32a**. Among the left and right arm bodies **32a**, the left arm body **32a** is configured to connect via a relay member **32c** to a lower connecting portion **21e** provided to the lower portion **21d** of the first exhaust-gas cleaning device **21**. The right arm body **32a** is configured to connect directly to the lower connecting portion **21e** of the first exhaust-gas cleaning device **21**.

(38) [Configuration of Position Adjustment of Exhaust-Gas Cleaning Devices]

(39) As illustrated in FIGS. **2**, **4**, and **5**, the support member **30** is provided with a position adjustment tool **35** that can adjust a support position—in the vehicle-body vertical direction, the vehicle-body longitudinal direction, and the vehicle-body lateral width direction—of the second exhaust-gas cleaning device **22** on the support column **31** and with a second position adjustment tool **37** that can adjust a support position—in the vehicle-body vertical direction, the vehicle-body longitudinal direction, and the vehicle-body lateral width direction—of the second exhaust-gas cleaning device **22** on the lower support portion **33**.

(40) When there is a position shift between the exhaust-gas discharging portion **21b** of the first exhaust-gas cleaning device **21** and the exhaust-gas introduction portion **22a** of the second exhaust-gas cleaning device **22**, so the positional relationship between the exhaust-gas discharging portion **21b** and the exhaust-gas introduction portion **22a** becomes appropriate and the exhaust-gas discharging portion **21b** and the exhaust-gas introduction portion **22a** can be appropriately connected by the connecting pipe **23** having no position adjustment function, the positional relationship between the exhaust-gas discharging portion **21b** and the exhaust-gas introduction portion **22a** can be adjusted by the position adjustment tool **35** and the second position adjustment tool **37**.

(41) Specifically, as illustrated in FIGS. **2**, **4**, and **5**, the position adjustment tool **35** is provided with an adjustment member **36** positioned between the front connection portion **22e** of the second exhaust-gas cleaning device **22** and the front support portion **31d** of the support column **31**, a first connecting bolt **36a** connecting the front connecting portion **22e** and the adjustment member **36**, and a second connecting bolt **36b** connecting the front support portion **31d** and the adjustment member **36**.

(42) As illustrated in FIGS. **2**, **4**, and **5**, the second position adjustment tool **37** is provided with a second adjustment member **38** positioned between the lower connecting portion **22f** of the second exhaust-gas cleaning device **22** and the lower support piece **33a** of the support column **31**, a third connecting bolt **38a** connecting the lower connecting portion **22f** and the second adjustment member **38**, and a fourth connecting bolt **38b** connecting the lower support piece **33a** and the second adjustment member **38**.

(43) A hole diameter of a through hole (not illustrated) of the adjustment member **36** into which the first connecting bolt **36a** is inserted is made greater than an outer diameter of the first connecting bolt **36a** for a configuration enabling position shifting of the front connecting portion **22e** relative to the adjustment member **36**. A through hole (not illustrated) of the position adjustment member **36** into which the second connecting bolt **36b** is inserted is made greater than an outer diameter of the second connecting bolt **36b** for a configuration enabling position shifting of the adjustment member **36** relative to the front support portion **31d**.

(44) A hole diameter of a through hole (not illustrated) of the second adjustment member **38** into which the third bolt **38a** is inserted is made greater than an outer diameter of the third connecting bolt **38a** for a configuration enabling position shifting of the adjustment member **36** relative to the front support portion **31d**. A hole diameter of a through hole (not illustrated) of the second adjustment member **38** into which the fourth connecting bolt **38b** is inserted is made greater than an outer diameter of the fourth connecting bolt **38b** for a configuration enabling position shifting of the adjustment member **36** relative to the front support portion **31d**.

(45) In the position adjustment tool **35**, among the first connecting bolt **36a**, the second connecting

bolt **36b**, the third connecting bolt **38a**, and the fourth connecting bolt **38b**, tightening of a connecting bolt corresponding to desired position adjustment is loosened. Performing an operation of moving the second exhaust-gas cleaning device **22** shifts the position of the second exhaust-gas cleaning device **22** relative to the support member **30**. This changes the support position—in the vehicle-body vertical direction, the vehicle-body longitudinal direction, and the vehicle-body lateral width direction—of the second exhaust-gas cleaning device **22** on the support column **31** in a manner corresponding to the operation that is performed of moving the second exhaust-gas cleaning device **22**.

(46) In the second position adjustment tool **37**, among the first connecting bolt **36a**, the second connecting bolt **36b**, the third connecting bolt **38a**, and the fourth connecting bolt **38b**, tightening of a connecting bolt corresponding to desired position adjustment is loosened. Performing an operation of moving the second exhaust-gas cleaning device **22** shifts the position of the second exhaust-gas cleaning device **22** relative to the support member **30**. This changes the support position—in the vehicle-body vertical direction, the vehicle-body longitudinal direction, and the vehicle-body lateral width direction—of the second exhaust-gas cleaning device **22** on the lower support portion **33** in a manner corresponding to the operation that is performed of moving the second exhaust-gas cleaning device **22**.

(47) [Configuration of Driving Unit] As illustrated in FIG. **1**, the driving unit **12** is provided behind the engine compartment **15**. As illustrated in FIGS. **1**, **3**, and **8**, the driving unit **12** is provided with the driver's seat **10**, the steering wheel **11** whereby the steering operation of the front wheels **1** is performed, a left brake operation unit **40L** whereby brakes (not illustrated) for the left back wheel are operated, and a right brake operation unit **40R** whereby brakes (not illustrated) for the right back wheel are operated.

[Configurations of Left Brake Operation Unit and Right Brake Operation Unit]

(48) As illustrated in FIG. **8**, the left brake operation unit **40L** and the right brake operation unit **40R** are provided to the right and below the steering wheel **11**. As illustrated in FIGS. **3**, **6**, and **8**, the left brake operation unit **40L** and the right brake operation unit **40R** are each provided with a brake pedal **41**, a return spring **42** that subjects the brake pedal **41** to an operation of being returned to an initial position (“off”), and a master cylinder **43** connected to the brake pedal **41**.

(49) As illustrated in FIG. **6**, the brake pedal **41** is provided with a pedal arm **41a** and with a footboard **41b** provided at a lower end portion of the pedal arm **41a**. A support shaft **44** is provided at an upper portion of the pedal arm **41a**. The support shaft **44** is supported by a support portion **45** provided to the traveling vehicle body **3**. The brake pedal **41** is supported on the traveling vehicle body **3** in a state of being able to swing between the initial position (“off”) and a braking position (“on”) by using a shaft core P of the support shaft **44** as a swinging fulcrum. The brake pedal **41** is positioned to the initial position (“off”) by the pedal arm **41a** abutting a stopper **47**.

(50) As illustrated in FIG. **6**, the partition member **17** forming the partition between the engine compartment **15** and the driving unit **12** is disposed further toward a vehicle-body front side than the brake pedal **41**. The return spring **42** is engaged to the pedal arm **41a** and a spring support portion **17d** provided to the partition member **17**. The return spring **42** biases the brake pedal **41** to swing to the initial position (“off”) by using the partition member **17** as a counterforce member.

(51) As illustrated in FIG. **6**, the partition member **17** is configured so the part **17a**, which corresponds to the footboard **41b**, of the partition member **17** is positioned further toward a vehicle-body front side than the part **17b**, which corresponds to the support shaft **44**, of the partition member **17**. A front portion of the footwell of the driving unit **12** can be extended forward past the support shaft **44**.

(52) As illustrated in FIG. **6**, the master cylinder **43** is provided on a vehicle-body back side of the support shaft **44** of the brake pedal **41**. The master cylinder **43** is fixed in a detachable manner to the support portion **45**. A slidable operation shaft **43a** of the master cylinder **43** and an operation arm **41c** provided to the brake pedal **41** are interconnected. The operation arm **41c** extends from the

pedal arm **41a** toward an opposite side of a side whereon the footboard **41b** is positioned relative to the support shaft **44**.

(53) In both the left brake operation unit **40L** and the right brake operation unit **40R**, when the brake pedal **41** is subjected to a stepping operation against the return spring **42** and enters the braking position (“on”), the operation shaft **43a** of the master cylinder **43** is subjected to a sliding operation to a pushed-in side by the operation arm **41c**. The master cylinder **43** supplies operational hydraulic pressure to the brakes (not illustrated), and the brakes perform an operation of switching to a braking state. When the stepping operation of the brake pedal **41** is released, the brake pedal **41** is subjected to an operation of returning to the initial position (“off”) by the return spring **42**, and the operation shaft **43a** of the master cylinder **43** is subjected to a sliding operation to a pulled-out side by the operation arm **41c**. The imparting of the operational hydraulic pressure by the master cylinder **43** to the brakes is released, and the brakes perform an operation of switching to an initial state.

(54) [Configuration of Return Spring]

(55) As illustrated in FIGS. **3** and **6**, the return spring **42** is constituted by a torsion coil spring and is provided with a coil portion **42a**, an arm **42b** extending from one end side of the coil portion **42a**, and an arm **42c** extending from another end side of the coil portion **42a**. The coil portion **42a** is fitted onto the support shaft **44**, and the return spring **42** is supported by the support shaft **44**. The arm **42b** on the one end side is engaged to the spring support portion **17d** of the partition member **17**. The arm **42c** on the other end side is engaged to the pedal arm **41a**.

(56) As illustrated in FIG. **6**, a bent end portion **42d** engaged to the pedal arm **41a** is provided to the arm **42c** on the other end side. A through hole **46** where to the bent end portion **42d** is engaged is provided to the pedal arm **41a**. A shape of the through hole **46** is made to be a shape into which the bent end portion **42d**, which moves along the shaft core P of the support shaft **44**, can be inserted. The return spring **42** and the brake pedal **41** can be assembled by the following assembly outline.

(57) As illustrated in FIG. **7**, the coil portion **42a** fits the return spring **42** onto the support shaft **44**, and the arm **42b** on the one end side is placed in a state of being engaged to the partition member **17**. The return spring **42** is in a free state. The stopper **47** is removed from the support portion **45**. The brake pedal **41** is attached to the support shaft **44** and swung up, and the brake pedal **41** is operated in a disposition opposing the support shaft, wherein the footboard **41b** is positioned in a position higher than when the brake pedal **41** is positioned in the initial position (“off”). The brake pedal **41** in this attitude of opposing the support shaft is guided to the support shaft **44** and moved toward the return spring **42**. By moving the brake pedal **41**, the bent end portion **42d** of the return spring **42** moves toward the pedal arm **41a** in a direction along the shaft core p of the support shaft **44**, is inserted into the through hole **46** from an inner side of the pedal arm **41a**, and moves to an outer side of the pedal arm **41a**. Next, the brake pedal **41** is swung down. At this time, the operation arm **41c** enters a slit **17e** provided in the partition member **17**. The stopper **47** is attached to the support portion **45**, and the brake pedal **41** is moved to the initial position (“off”). This causes the brake pedal **41** to be stopped by the stopper **47**. As illustrated in FIG. **6**, the bent end portion **42d** engages to the pedal arm **41a**, the return spring **42** is provided with an elastic restoring force, and the brake pedal **41** enters a state of being biased and swung to the initial position (“off”) by the return spring **42**. The slit **17e** is closed off once assembly of the brake pedal **41** is finished.

(58) [Configurations of Steering-Post Cover and Panel Cover]

(59) The steering wheel **11** is supported in a state of being able to change positions in the vehicle-body longitudinal direction and the vehicle-body vertical direction. As illustrated in FIG. **8**, a steering-post cover **50** is inserted into a through hole **52** of a panel cover **51** supported on the traveling vehicle body **3**. When the steering wheel **11** changes positions in the vehicle-body longitudinal direction, the steering-post cover **50** follows the steering wheel **11** and swings in the vehicle-body longitudinal direction relative to the panel cover **51**. A bellows-shaped expandable cover **53** is connected across a proximal-side lower portion of the steering-post cover **50** and a

lower portion **51a** of the panel cover **51**. The expandable cover **53** expands and contracts according to the swinging of the steering-post cover **50** but, regardless of the swinging of the steering-post cover **50**, places a gap between the steering-post cover **50** and the lower portion **51a** of the panel cover **51** in a closed state. Between a part opposing a distal-side wall portion of the steering-post cover **50** of the panel cover **51** and the distal-side wall portion of the steering-post cover **50**, a gap that allows the steering-post cover **50** to swing is provided. However, above the gap and on the distal-side wall portion of the steering-post cover **50**, a bulging portion that bulges toward a cover outer side and makes the gap difficult to see from above is provided. In the panel cover **51**, a face portion **51b** corresponding to a side portion of the steering-post cover **50** is raised to the proximal side. Even if the steering-post cover **50** swings, the face portion **51b** makes it difficult to see the gap between the steering-post cover **50** and the panel cover **51** on a horizontal side of the steering-post cover **50**.

Other Embodiments

(60) (1) FIG. **9** is a back view of a panel cover **55** provided with another example of one or more embodiments. FIG. **10** is a perspective view of a steering post cover **56** provided with another example of one or more embodiments. In the panel cover **55** provided with another example, below a through hole **57** into which the steering-post cover **56** is inserted, an opening **58** and a proximal wall **59** extending upward from a lower edge portion of the opening **58** are provided. In the steering-post cover **56**, on a proximal-side lower portion, a proximal extended portion **61** that extends lower than a lower end **60a** of a horizontal wall portion **60** is provided, and on a distal-side lower portion, a distal extended portion **62** that extends lower than the lower end **60a** of the horizontal wall portion **60** is provided. The proximal wall **59**, the proximal extended portion **61**, and the distal extended portion **62** serve as hiding walls that make it difficult to see into an area below the panel cover **55** via a gap arising between the steering-post cover **56** and the panel cover **55** on a proximal side and horizontal side of the steering-post cover **56**.

(61) (2) In the embodiments described above, a configuration in which the support column **31** extends from the flywheel housing **6** is given, but the configuration is not limited thereto, and the support column **31** may extend from any portion of the vehicle body frame **4**.

(62) (3) In the embodiments described above, an example in which the position adjustment tool **35** is provided is given, but the position adjustment tool **35** need not be provided.

(63) (4) In the embodiments described above, an example in which the lower support portion **33** and the second position adjustment tool **37** is provided is given, but the lower support portion **33** need not be provided. Also, the lower support portion **33** may be provided, and the second position adjustment tool **37** need not be provided.

(64) (5) In the embodiments described above, an example in which the lower support portion **33** is provided on the washer portion **34** is given, but the lower support portion **33** need not be provided on the washer portion **34** and may be provided directly on the support column **31**.

INDUSTRIAL APPLICABILITY

(65) One or more embodiments of the present invention can be applied to a tractor provided with an engine bonnet forming an engine compartment.

(66) Although the disclosure has been described with respect to only a limited number of embodiments, those skilled in the art, having benefit of this disclosure, will appreciate that various other embodiments may be devised without departing from the scope of the present invention. Accordingly, the scope of the invention should be limited only by the attached claims.

Claims

1. A tractor comprising: a vehicle body; an engine bonnet that forms an engine compartment; an engine that is disposed in the engine compartment; a first exhaust-gas cleaning device that is disposed in the engine compartment and purifies exhaust gas discharged from the engine, the first

exhaust-gas cleaning device being disposed above the engine such that a long side of the first exhaust-gas cleaning device is substantially parallel to a vehicle-body lateral width direction; a second exhaust-gas cleaning device that is disposed in the engine compartment and purifies the exhaust gas, the second exhaust-gas cleaning device being disposed behind the engine such that a long side of the second exhaust-gas cleaning device is substantially parallel to the vehicle-body lateral width direction; a vehicle body frame; and a supporting tool that: is supported by the vehicle body frame, supports the first exhaust-gas cleaning device and the second exhaust-gas cleaning device, and comprises: a support column that extends upward from the vehicle body frame along a front portion of the second exhaust-gas cleaning device while supporting the front portion, and a support arm that extends forward from an upper portion of the support column along a lower portion of the first exhaust-gas cleaning device while supporting the lower portion, wherein the vehicle body frame comprises: a flywheel housing that extends backward from a back portion of the engine and positioned below the second exhaust-gas cleaning device, and the support column extends upward from the flywheel housing.

2. The tractor according to claim 1, further comprising: a first position adjustment tool that adjusts a support position of the second exhaust-gas cleaning device, on the support column, in a vehicle-body vertical direction, a vehicle-body longitudinal direction, and a vehicle-body lateral width direction.

3. The tractor according to claim 1, further comprising: a lower support portion that is supported below the support column and supports a lower portion of the second exhaust-gas cleaning device.

4. The tractor according to claim 3, further comprising: a second position adjustment tool that adjusts a support position of the second exhaust-gas cleaning device, on the lower support portion, in a vehicle-body vertical direction, a vehicle-body longitudinal direction, and a vehicle-body lateral width direction.

5. A tractor comprising: a vehicle body; an engine bonnet that forms an engine compartment; an engine that is disposed in the engine compartment; a first exhaust-gas cleaning device that is disposed in the engine compartment and purifies exhaust gas discharged from the engine, the first exhaust-gas cleaning device being disposed above the engine such that a long side of the first exhaust-gas cleaning device is substantially parallel to a vehicle-body lateral width direction; a second exhaust-gas cleaning device that is disposed in the engine compartment and purifies the exhaust gas, the second exhaust-gas cleaning device being disposed behind the engine such that a long side of the second exhaust-gas cleaning device is substantially parallel to the vehicle-body lateral width direction; a vehicle body frame; a supporting tool that: is supported by the vehicle body frame, supports the first exhaust-gas cleaning device and the second exhaust-gas cleaning device, and comprises: a support column that extends upward from the vehicle body frame along a front portion of the second exhaust-gas cleaning device while supporting the front portion, and a support arm that extends forward from an upper portion of the support column along a lower portion of the first exhaust-gas cleaning device while supporting the lower portion; and a lower support portion that is supported below the support column and supports a lower portion of the second exhaust-gas cleaning device, wherein the support column comprises: a washer portion that is disposed at a lower portion of the support column and is connected to the vehicle body frame to make the support column supported by the vehicle body frame, and the lower support portion constitutes a part of the washer portion.
